Chest X-Ray Pneumonia Detection using ResNet50 Transfer Learning

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Summary
A binary chest X-ray classifier (NORMAL vs. PNEUMONIA) built on ResNet50 transfer learning, served via a FastAPI REST API, and containerized with Docker. The model achieves **AUC-ROC = 0.9576** and **92% accuracy** at the tuned threshold of 0.87, with PNEUMONIA recall of **96%** and NORMAL recall of **85%**. Live API tests confirm correct predictions: NORMAL at 100% confidence and PNEUMONIA at 96.9% confidence.

1. Problem Definition

Pneumonia is responsible for approximately **2.5 million deaths per year** globally, making it the single largest infectious cause of death in children under five. Early and accurate diagnosis is critical, yet manual interpretation of chest X-rays is slow, subjective, and heavily dependent on radiologist availability — a resource that is scarce in many parts of the world.

This project addresses the problem as a **binary image classification task**: given a posteroanterior (PA) chest X-ray image, classify it as either **NORMAL** or **PNEUMONIA**. The system is designed as an *AI-assisted decision-support tool* for radiologists and clinicians — not a replacement. It flags suspicious cases for priority review, reducing cognitive load and enabling faster triage.

Task Specification

Parameter	Specification
Intended user	Radiologists, clinicians, emergency triage staff
Input	JPEG or PNG chest X-ray (any resolution — resized to 224×224)
Output	{prediction, confidence, prob_pneumonia, prob_normal, threshold_used, inference_time_ms}
Classes	NORMAL | PNEUMONIA (binary)
Threshold	0.87 (tuned); configurable per-request

Success Criteria

- Overall accuracy > 90% on the held-out test set
- PNEUMONIA recall (sensitivity) > 95% — minimising missed diagnoses
- AUC-ROC > 0.95 — strong discriminative ability across all thresholds

Why AI? The volume of chest X-rays produced in hospitals far exceeds the capacity for timely manual review. An AI system provides an instant second opinion, consistent scoring criteria, and can help prioritise which patients need urgent radiologist attention.

2. Dataset and Data Strategy

The model was trained and evaluated on the **Kaggle Chest X-Ray Images (Pneumonia)** dataset (Kermany et al., 2018). Source: https://www.kaggle.com/paultimothymooney/chest-xray-pneumonia

Dataset Statistics

Split	NORMAL	PNEUMONIA	Total
Train	1,349	3,883	5,232
Test	234	390	624
Total	1,583	4,273	5,856

The dataset has a **1:2.9 class imbalance** (NORMAL:PNEUMONIA). The original validation split (16 images) was too small to be statistically meaningful, so it was merged back into the training set, with

an 80/20 stratified split used instead.

Preprocessing

• Images decoded with OpenCV (cv2.imdecode), converted from BGR to RGB, and resized to 224×224 pixels.

• Pixel normalisation uses ResNet50's **preprocess_input** function, which subtracts ImageNet BGR channel means [103.939, 116.779, 123.68]. This is NOT a simple /255 rescale — using the correct function is essential for transfer learning to work correctly.

• No augmentation is applied to the validation or test sets.

Data Augmentation Strategy

Class-specific augmentation is used to compensate for the 1:2.9 imbalance:

• NORMAL (minority class): stronger augmentation — rotation ±15°, zoom 15%, width/height shifts, shear, horizontal flip.

• PNEUMONIA: lighter augmentation — rotation ±10°, zoom 10%, horizontal flip.

• Balanced batch generator yields equal batches from each class with sample weights baked in (Keras 3 does not support class_weight with Python generators).

• Data leakage prevention: test set is completely held out throughout training.

3. Model Architecture and Training

The model uses **transfer learning** with ResNet50 (He et al., 2015) pretrained on ImageNet as a feature extractor backbone. ResNet50 contains approximately 23.6M parameters across 50 layers with residual connections that enable very deep network training without vanishing gradients.

Architecture

Layer	Output Shape	Parameters
ResNet50 backbone	(7, 7, 2048)	~23.6M (frozen phase 1)
GlobalAveragePooling2D	(2048,)	0
BatchNormalization	(2048,)	8,192
Dense(256, ReLU)	(256,)	524,544
Dropout(0.4)	(256,)	0
Dense(64, ReLU)	(64,)	16,448
Dropout(0.2)	(64,)	0
Dense(1, Sigmoid)	(1,)	65

Two-Phase Training Strategy

Phase 1 — Head Training (10 epochs): ResNet50 base is fully frozen. Only the custom classification head is trained with Adam at lr=1e-3. Val AUC reached **0.9971**. This phase stabilises the new layers before any disturbance to pretrained features.

Phase 2 — Fine-Tuning (15 epochs): The top 30 layers of ResNet50 (layers 146–175) are unfrozen. The entire model is trained end-to-end with Adam at lr=1e-4 to prevent catastrophic forgetting. Val AUC reached **0.9913**.

Training Configuration

Setting	Value
Framework	TensorFlow 2.21 / Keras 3
Loss	Binary cross-entropy with sample weights
Optimizer	Adam (lr=1e-3 phase 1, lr=1e-4 phase 2)
EarlyStopping	monitor=val_auc, patience=5, restore_best_weights=True
ReduceLROnPlateau	factor=0.3, patience=3, monitor=val_auc
Input size	224 × 224 × 3
Batch size	32 (balanced: 16 NORMAL + 16 PNEUMONIA)

4. Evaluation Results

The model is evaluated on the held-out test set of 624 images. The AUC-ROC of **0.9576** exceeds the target of 0.95, confirming strong discriminative ability. Threshold tuning dramatically improves clinical utility compared to the default sigmoid output of 0.5.

Table 1: Performance at Different Thresholds

Metric	Threshold = 0.5 (default)	Threshold = 0.87 (optimal)
Accuracy	78%	92%
AUC-ROC	0.9576	0.9576
PNEUMONIA Precision	74%	90%
PNEUMONIA Recall (Sens.)	99%	96%
NORMAL Precision	98%	96%
NORMAL Recall (Spec.)	41%	85%
F1 — PNEUMONIA	0.85	0.93
F1 — NORMAL	0.58	0.90

Confusion Matrix at Threshold = 0.87 (test set, 624 images)

	Predicted NORMAL	Predicted PNEUMONIA
Actual NORMAL	TN = 199	FP = 35
Actual PNEUMONIA	FN = 16	TP = 374

Live API Test Results

The following results were obtained from live API testing of the deployed system:

Test	Prediction	Confidence	prob_pneumonia	Inference Time	Status
NORMAL X-ray	NORMAL	100.0%	0.01%	149 ms	Correct
PNEUMONIA X-ray	PNEUMONIA	96.9%	96.9%	700 ms	Correct

The longer inference time for the PNEUMONIA case (700ms) is because the /predict/explain endpoint was used, which includes Grad-CAM heatmap generation. The fast /predict endpoint consistently achieves ~150ms.

Key finding: Threshold tuning from 0.5 to 0.87 improved NORMAL recall (specificity) from 41% to 85% while keeping PNEUMONIA recall at 96% — a major clinical improvement with only a minor trade-off in sensitivity.

5. Error Analysis and Failure Cases

Understanding failure modes is critical before any clinical deployment. At threshold 0.87, the model makes 51 errors on the 624-image test set: 35 False Positives and 16 False Negatives.

False Positives (35 cases — NORMAL predicted as PNEUMONIA)

Normal chest X-rays classified as pneumonia. Common causes include subtle opacities, atelectasis, pleural effusion, or image artefacts that superficially resemble early pneumonia patterns. **Clinical impact:** unnecessary antibiotic treatment, patient anxiety, and added diagnostic costs. These are undesirable but not immediately life-threatening.

False Negatives (16 cases — PNEUMONIA missed)

Pneumonia cases missed by the model. **These are the most dangerous errors** in a clinical context. They typically involve subtle early-stage consolidation, opacities obscured by overlying structures, or atypical presentations. The threshold was specifically tuned to 0.87 (from 0.5) to minimise these cases — at threshold 0.5, the raw sigmoid output produces 41% NORMAL recall (many FPs), but raising the threshold to 0.87 dramatically reduces FPs while PNEUMONIA recall only drops from 99% to 96%.

Grad-CAM Analysis

Gradient-weighted Class Activation Mapping (Grad-CAM) was used to validate that the model attends to clinically relevant regions:

- PNEUMONIA cases: activations concentrate in the lower lung lobes — the typical anatomical location of bacterial and viral pneumonia consolidation.
- NORMAL cases: more diffuse activation spread across both lung fields with no single dominant hot region — consistent with the absence of focal pathology.
- No activations on cardiac silhouette, clavicles, or image borders — confirming the model uses anatomically appropriate features.

Known Limitations

- Single dataset from one institution — may not generalise to different equipment or populations.
- No demographic diversity analysis performed (age, sex, ethnicity).
- Trained only on bacterial and viral pneumonia; not validated for COVID-19 or fungal pneumonia.
- Not validated on clinical-grade PACS/DICOM images; tested only on JPEG/PNG exports.
- Should not be used as a standalone diagnostic tool without radiologist oversight.

6. System Architecture and Inference Pipeline

Inference Pipeline

```
Input X-Ray (JPEG/PNG)
↓
Preprocessing: cv2.imdecode → resize(224×224) → BGR2RGB → preprocess_input
↓
ResNet50 Backbone (frozen conv layers 1–145, fine-tuned layers 146–175)
↓
Custom Head: GAP → BatchNorm → Dense(256) → Dropout → Dense(64) → Dropout → Sigmoid
↓
Probability score [0.0, 1.0]
↓
Threshold (0.87): prob ≥ 0.87 → PNEUMONIA, prob < 0.87 → NORMAL
↓
Output: {prediction, confidence, prob_pneumonia, prob_normal,
threshold_used, inference_time_ms}
```

API Endpoints

Method	Endpoint	Description
GET	/health	Server status and active threshold
GET	/model/info	Model metadata (path, input size, classes)
POST	/predict	Single image — ~150ms inference
POST	/predict/explain	Classification + Grad-CAM heatmap (base64 PNG) — ~700ms

Infrastructure

- Framework: FastAPI served via Uvicorn — async, non-blocking request handling
- Containerisation: Docker + docker-compose — runs identically on any device
- Model load: once at startup; TensorFlow session reused for subsequent requests
- Web UI: localhost:8000 — drag-and-drop interface with Single Image and Batch tabs

- Inference time: ~150ms (/predict) or ~700–900ms (/predict/explain with Grad-CAM)

7. Bonus Implementations

Feature	Description	Location
Docker	Full containerisation — docker build + docker run -p 8000:8000	Dockerfile, docker-compose.yml
Grad-CAM	Heatmap overlay showing which lung regions the model attended to	src/inference/grad_cam.py
Web UI	localhost:8000 — Single Image tab (result + Grad-CAM) and Batch tab (summary + expandable cards	src/api/static/index.html
ONNX Export	scripts/export_onnx.py exports model for edge/cross-platform deployment	scripts/export_onnx.py
Automated Tests	pytest suite covering all endpoints with real image fixtures	tests/test_api.py
Threshold Tuning	F1-score sweep to find optimal threshold (0.87 vs default 0.5)	scripts/evaluate.py